Juniper Networks® vMX Release Notes

Release 15.1F3
22 July 2020
Revision 1

These release notes accompany this release of the Juniper Networks® virtual MX Series router (vMX). They describe features, limitations, and known problems in the software.

Contents

Introduction

The virtual MX Series router (vMX) is an MX Series router optimized to run on x86 servers. We recommend Ubuntu as the host OS.

vMX supports most of the features available on MX Series routers and allows you to leverage Junos OS Release 15.1 to provide quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations by consistency with MX Series routers
- Introduces new services without reconfiguration of current infrastructure

Features Supported in This Release

This release of vMX supports most of the features available on Juniper Networks MX Series routers with the following exception:

- High availability features such as virtual Routing Engine redundancy are not supported in this release.

Licensing

Licenses are required for using vMX features. When you order licenses, this information is bound to a customer ID. If you did not order the licenses, contact your account team or Juniper Networks Customer Care for assistance. When you order a license, you receive instructions for generating license activation keys on the Juniper Networks License Management System.

The vMX licenses are based on application packages and processing capacity. Table 1 on page 2 describes the features available with application packages.

Table 1: Application Packages for Licenses

Application Package	Features
BASE	IP routing with 32,000 routes in the forwarding table Basic Layer 2 functionality, Layer 2 bridging and switching

Table 1: Application Packages for Licenses *(continued)*

Application Package	Features
ADVANCE	Features in the BASE application package IP routing with routes up to platform scale in the forwarding table IP and MPLS switching for unicast and multicast applications Layer 2 features include Layer 2 VPN, VPLS, EVPN, and Layer 2 Circuit VXLAN
PREMIUM	Features in the BASE and ADVANCE application packages Layer 3 VPN for IP and multicast

An application package is associated with a bandwidth license. vMX provides full-duplex bandwidth in the following capacities: 100 Mbps, 250 Mbps, 500 Mbps, 1 Gbps, 5 Gbps, 10 Gbps, and 40 Gbps. Bandwidth licenses that are not associated with a specific application package apply to all application packages.

You can download the vMX software BASE application package with 1 Mbps bandwidth and evaluate it for 30 days without a license. To use additional features beyond the 30 days, you must order the appropriate license.

Minimum Hardware and Software Requirements

Table 2 on page 3 lists the hardware requirements.

Table 2: Minimum Hardware Requirements

Description	Value
Sample system configuration	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability can be used. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache For single root I/O virtualization (SR-IOV) NIC type, use Intel 82599-based PCI-Express cards (10 Gbps) and Ivy Bridge processors.

Table 2: Minimum Hardware Requirements *(continued)*

Description	Value
Number of cores	For lab simulation use case: 2 (1 for VCP and 1 for VFP) For low-bandwidth applications: 4 (1 for VCP and 3 for VFP) For high-bandwidth applications: 8 (1 for VCP and 7 for VFP)
Memory	For lab simulation use case or low-bandwidth applications: Minimum of 10 GB (2 GB for VCP, 8 GB for VFP) For high-bandwidth applications: Minimum of 14 GB (2 GB for VCP, 12 GB for VFP) Additional 2 GB recommended for host OS
Storage	Local or NAS
Other requirements	Intel VT-d capability

Table 3 on page 4 lists the software requirements.

Table 3: Minimum Software Requirements

Description	Value
Operating system	Ubuntu 14.04 LTS (recommended host OS) Linux 3.13.0-32-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11 or later
Required packages **NOTE:** Other additional packages might be required to satisfy all dependencies.	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev **NOTE:** libvirt 1.2.8

Known Behavior

This section contains the known behaviors and limitations in this release.

- Scale limitation is observed with VLAN tag operation and circuit cross-connect (CCC).

Known Issues

This section lists the known issues in this release.

- vMX traffic is not restored until 140 seconds after the VFP installs the routes. PR1011480
- vMX stops sending ICMPv6 echo-reply messages when receiving 100 Kpps ICMPv6 echo-request messages. PR1053964
- LACP packets are getting dropped on the bridge. PR1059231
- Installing vMX after a server reboot fails the first time because of a HugePage error (for virtio). PR1060438
- LLDP packets are getting dropped on the bridge (for virtio). PR1066850
- After an FPC restart, **/kernel**, **chassisd**, and **i386_junos** error messages appear in the syslog file. PR1070585
- When the FPC is restarting, **COS(cos_chassis_scheduler_pre_add_action:2137)** error messages appear indicating that flexible queuing mode is not enabled. PR1070655
- Output for the **show pfe statistics traffic** command might take a few seconds to appear and many **MIB2D_COUNTER_DECREASING: pfes_stats_delta: counter** messages appear in the log file before the response time improves. PR1071659
- When the FPC is restarting, a **Received unsupported pic_mask 0x1 ignored message** message appears in the syslog file. PR1072436
- Traffic loss occurs at a remote receiver because of lost remote PIM joins to the local receiver. PR1087031
- Using ssh to connect to the VFP might have a noticeable delay. PR1089935

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or JNASC support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf .
- Product warranties—For product warranty information, visit https://www.juniper.net/support/warranty/ .
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: https://www.juniper.net/customers/support/
- Search for known bugs: http://www2.juniper.net/kb/
- Find product documentation: https://www.juniper.net/documentation/
- Find solutions and answer questions using our Knowledge Base: https://kb.juniper.net/
- Download the latest versions of software and review release notes: https://www.juniper.net/customers/csc/software/
- Search technical bulletins for relevant hardware and software notifications: https://www.juniper.net/alerts/
- Join and participate in the Juniper Networks Community Forum: https://www.juniper.net/company/communities/
- Open a case online in the CSC Case Management tool: https://www.juniper.net/cm/

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: https://tools.juniper.net/SerialNumberEntitlementSearch/

Opening a Case with JTAC

You can open a case with JTAC on the Web or by telephone.

- Use the Case Management tool in the CSC at https://www.juniper.net/cm/ .
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see https://www.juniper.net/support/requesting-support.html .

Revision History

22 July 2020—Revision 2, Junos OS Release 15.1F3—vMX

30 October 2015—Revision 1, Junos OS Release 15.1F3—vMX